



Changes in a seagrass-associated decapod assemblage after 25 years of eutrophication and degradation of a Caribbean reef lagoon

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Abstract

Coastal eutrophication and habitat degradation can have long-term impacts on coastal ecosystems and communities. In the Mexican Caribbean, the Puerto Morelos reef lagoon was highly oligotrophic in 1995, but by 2020 it had changed to mesotrophic-eutrophic because of a gradual increase in nutrient and organic matter inputs related with an explosive growth in population and coastal development since the late 1990s, and massive inundations of holopelagic *Sargassum* since 2014. Here, we aimed to compare the seagrass-associated decapod assemblage in the reef lagoon after 25 years of environmental change. In summer and winter of both periods (1995 and 2020), day and night samples were taken in the same way on five lagoon sites. Of 159 decapod taxa, 108 were obtained in 2020 and 106 in 1995. The number of decapods collected in 2020 (138,046) was 2.5 times the number collected in 1995 (53,211). Three canopy-dwelling carideans (*Latreutes fucorum*, *Palaemonella americana*, and *Thor* spp.) accounted for ~50% of individuals in 1995, whereas bottom-dwelling hermit crabs *Clibanarius tricolor* and *Pagurus* spp. constituted 56% of individuals in 2020. Richness and abundance were higher in 2020 and night samples, but did not vary with season. The assemblage composition also varied more with period and time of day than with season. These findings, and the change in dominance from omnivore/herbivore shrimps in 1995 to omnivore/detritivore hermit crabs in 2020, suggest increases in benthic eutrophication over time. Our study provides insights into how ecological changes are affecting seagrass-associated invertebrates.

Keywords Caridean shrimps · Hermit crabs · Assemblage composition · Reef lagoon · *Sargassum* inundations

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Introduction

In tropical zones, seagrass meadows are among the most diverse and ecologically significant shallow-water marine ecosystems (Nordlund et al. 2016; York et al. 2018). Seagrasses influence nutrient cycling and provide critical habitat for numerous fish and invertebrate species (Du et al. 2020). They also function as foraging grounds for some reef-associated fauna, including many species of importance to fisheries, and as nursery areas for juveniles of species that migrate to coral reefs as they grow (Duffy 2006; Harborne et al. 2006). Seagrasses are vulnerable to climate change and other anthropogenic stressors, such as decreases in water quality, eutrophication, sedimentation, fishing activities, and the introduction of non-native species (Burkholder et al. 2007; Hughes et al. 2009; York et al. 2018). Seagrass ecosystems in the wider Caribbean region, in particular, have sustained high rates of degradation (Duffy 2006; Arias-Schriber et al. 2008; van Tussenbroek et al. 2014; UNEP-CEP 2021; Velázquez-Ochoa and Enríquez 2023; Brewton and Lapointe 2025), potentially affecting their associated communities.

During the last few decades, dramatic coastal development and exponential growth of local populations and tourist visitors have taken place on the Mexican Caribbean coast along the state of Quintana Roo (i.e., the eastern margin of the Yucatan Peninsula), particularly along its northern portion, where the cities of Cancún and Puerto Morelos are located (Suchley and Álvarez-Filip 2018). In Cancún, for example, between 1995 and 2021, the number of hotel rooms increased from 18,000 to 40,000 and the number of visitors from about 2 to 9 million (INEGI 2004, 2022). Similarly, between 1995 and 2021, the population of Puerto Morelos increased from ~1,500 to 31,500 and the number of hotel rooms from 400 to 7,000 (Rodríguez-Martínez et al. 2010; INEGI 2022; SEDETURQROO 2025). Because the Yucatan Peninsula is a karstic region with an underwater circulation system that discharges in the sea through underwater springs, and sewage management and treatment is primary at best, this exponential growth has substantially increased the input of nutrients and contaminants into coastal waters and sediments (Rodríguez-Martínez et al. 2010; Hernández-Terrones et al. 2015; Suchley and Álvarez-Filip 2018; Velázquez-Ochoa and Enríquez 2023).

Since the mid-1990s, gradual changes have been occurring in the seagrass meadows of the Puerto Morelos reef lagoon. The abundance of fast-growing species such as the manatee seagrass *Syringodium filiforme* as well as fleshy and drift algae has been increasing. A greater allocation to above-ground than to below-ground biomass by the foundation species, the turtle grass *Thalassia testudinum*, and an increase in the phosphorus content in the leaves of *T. testudinum* have been recorded (Carruthers et al. 2005; van Tussenbroek 2011; Caballero-Aragón et al. 2022). Anthropogenic sources of nitrogen, such as sewage pollution, are reflected in increases in the leaf area index and in the values of the nitrogen stable isotope ($\delta^{15}\text{N}$) in *T. testudinum* leaves (Velázquez-Ochoa and Enríquez 2023). These changes are consistent with increased nutrient input into the reef lagoon.

In addition to these anthropogenic sources, unprecedented massive inundations of holopelagic brown algae *Sargassum* spp. have been arriving on the Mexican Caribbean coast since 2014 (van Tussenbroek et al. 2017; Cabanillas-Terán et al. 2019; Chávez et al. 2020). Upon reaching the coastline, thousands of tons of *Sargassum* become stranded and rapidly decay, turning the coastal waters brown and turbid (Online Resource 1). These “*Sargassum* brown tides” further increase the nutrient input and exacerbate the degradation of the coastal

seagrass meadows (van Tussenbroek et al. 2017; Velázquez-Ochoa and Enríquez 2023), potentially affecting the seagrass-associated fauna (Rodríguez-Martínez et al. 2019).

Several studies have addressed the effects of coastal development and the *Sargassum* brown tides on the water and sediment quality, as well as on the vegetation of the reef lagoon (e.g., van Tussenbroek et al. 2017; Chávez et al. 2020; Rodríguez-Muñoz et al. 2021; Velázquez-Ochoa and Enríquez 2023). Yet, their effects on the motile invertebrate fauna living on seagrass leaves and on the adjacent bottom have received less attention. This fauna constitutes a large portion of the biodiversity in seagrass systems (Boström et al. 2006; Clarke et al. 2017; York et al. 2018; Hyman et al. 2019), in particular crustaceans, which are important links in seagrass trophic webs (Heck and Orth 1980; Gore et al. 1981; García-Raso et al. 2006; Boada et al. 2018; Mariño et al. 2018).

Long-term data series may be the best approach to assess ecological changes and variability over time, but are rarely available (Southward 1995; Magurran et al. 2010; García-Barón et al. 2021). In their absence, baseline surveys (i.e., surveys conducted before the changes occurred) and resampling of the same sites surveyed in the past can be valuable tools for assessing how environmental changes affect the diversity and composition of faunal communities (e.g., Bianchi et al. 2014; Callaway 2016; Schmitter-Soto et al. 2018; Kendall et al. 2021; Zampa et al. 2023). The fauna of decapod crustaceans associated with seagrasses in the Puerto Morelos reef lagoon was characterized in 1995/98, before the explosive coastal development and well before the onset of the massive *Sargassum* influxes (Monroy-Velázquez 2000; Briones-Fourzán et al. 2020). Back then, conditions in the reef lagoon were oligotrophic, the sediments were dominated by coarse carbonate sediments with low organic content (Enríquez et al. 2001), and *T. testudinum*, the dominant seagrass, had more below- than above-ground biomass (van Tussenbroek 1995; van Tussenbroek 1998). By 2020, however, sedimentation rates had increased considerably, sediments had undergone a substantial enrichment in organic matter and ammonium, and the conditions in the reef lagoon had become mesotrophic-eutrophic (López-Mendoza et al. 2020; Pérez-Gómez et al. 2020; Velázquez-Ochoa and Enríquez 2023).

Surveys conducted in other seagrass ecosystems suggest that the seagrass-associated crustacean fauna is resilient to ecosystem changes related to coastal development (e.g., Matheson et al. 1999; Blake et al. 2014; Brewton and Lapointe 2025), probably because this fauna contains many small-sized species with fast reproductive cycles and a short life span (Gartner et al. 2015). However, if environmental changes such as eutrophication and sedimentation increase gradually, it may take many years to detect their potential effects on this fauna (Heip 1995). The time elapsed since the baseline survey presented an opportunity to assess changes in ecological indices and assemblage composition of the seagrass-associated decapod assemblage after substantial environmental changes. Therefore, the aim of the present study was to conduct such an assessment after the 25-year gap between the baseline survey and a resampling conducted in 2020. Based on changes documented in the Puerto Morelos reef lagoon and seagrass meadows since 1995 and the resilience of this fauna, we expected some level of species turnover and changes in species dominance and abundance, but not necessarily substantial changes in decapod assemblage composition.

Materials and methods

Study area

The Puerto Morelos reef lagoon ($20^{\circ}48' \text{ N}$, $86^{\circ} 51' \text{ W}$) is a shallow ($\leq 5 \text{ m}$ in depth) body of water with an area of $\sim 43.4 \text{ km}^2$ that extends from the shoreline to a series of coral reefs that run slightly diagonally to the coast, at a distance of 500–1500 m (Fig. 1). In 1998, the reef lagoon and coral reefs (i.e., the coral reef system) were declared a national park (Puerto Morelos Reef National Park). At the time of the baseline survey, conditions in the reef lagoon were oligotrophic, with low levels of nitrite, nitrate, and phosphate in both water and porewater (Duarte et al. 1995; Ruiz-Rentería et al. 1998). The bottom of the lagoon was covered by a well-developed seagrass community dominated by *T. testudinum*, with *S. filiforme* and a variety of rhizophytic macroalgae growing interspersed with *T. testudinum* (van Tussenbroek 1995).

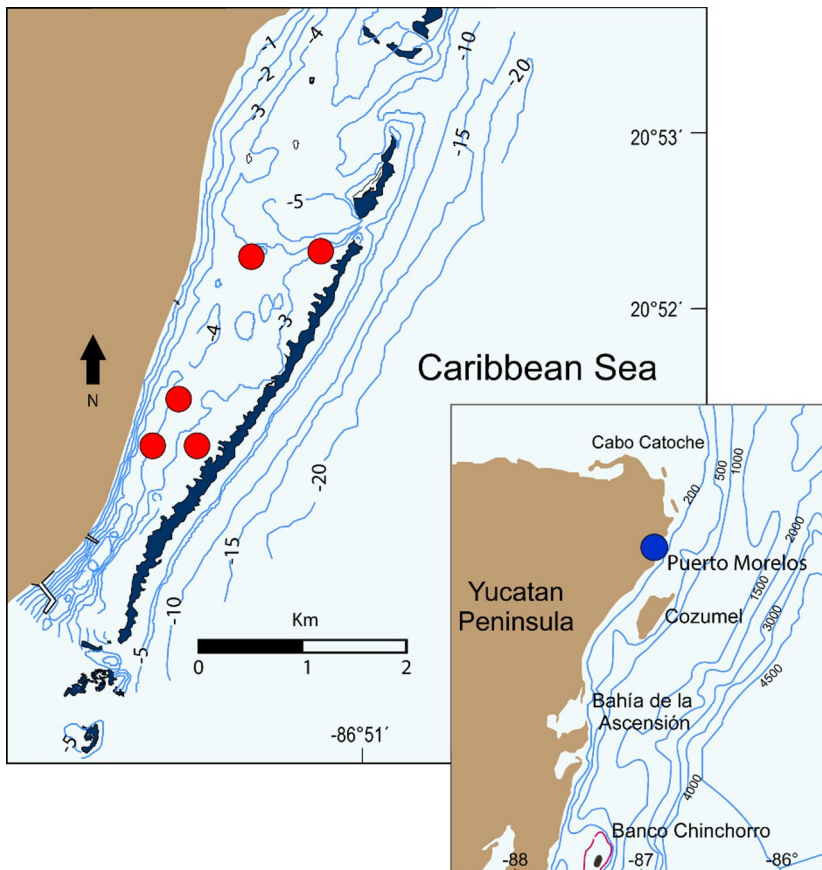


Fig. 1 Schematic map of the study area (Puerto Morelos reef lagoon, Mexico). Red dots indicate the five sampling sites, and black areas represent the reef crest. The inset shows the location of Puerto Morelos on the Mexican Caribbean coast (blue dot). Isobaths are in meters

By 2020, however, although *T. testudinum* was the overall dominant seagrass, *S. filiforme* or the shoal grass *Halodule wrightii* dominated in some lagoon areas (Caballero-Aragón et al. 2022), and the conditions in the reef lagoon were considered mesotrophic-eutrophic, with high levels of water and porewater nutrients due to exposure to anthropogenic pressure, resulting in dense seagrass canopies (Velázquez-Ochoa and Enríquez 2023). Groundwater discharges have been identified as the primary source of nutrient enrichment (Null et al. 2014; Hernández-Terrones et al. 2015; Pérez-Gómez et al. 2020).

Epifaunal sampling

The baseline survey was conducted in June–July 1995 (summer) and February–March 1998 (winter). Because no differences in decapod abundance, species richness, or community composition were found between these samplings (Briones-Fourzán et al. 2020), we hereafter refer to the baseline period simply as “1995”. In both 1995 and 2020, epifauna samples were collected by the same researchers using the same method at the same five seagrass sites in the southern half of the reef lagoon (Fig. 1).

On each site, the same epibenthic sled (mouth aperture: 0.57 m wide · 0.25 m high, mesh aperture: 1 mm, length: 1.19 m) was trawled 10 times during the day (1100–1300 h) and 10 times during the night (2000–2200 h). Each trawl lasted for 1 min at a speed of 1 m s⁻¹. The average trawl length was 60 m; therefore, the 10 trawls sampled an area of 342 m² on average (Briones-Fourzán and Lozano-Álvarez 2001; Briones-Fourzán et al. 2020). In 1995, there were some differences in the vegetation between the back-reef zone (a 100–400 m-wide strip adjacent to the coral reefs) and the mid-lagoon zone (the greater part of the lagoon) (Ruiz-Rentería et al. 1998; Briones-Fourzán and Lozano-Álvarez 2001), but no effect of lagoon zone was found on the decapod richness, abundance, or community composition (Monroy-Velázquez 2000; Briones-Fourzán et al. 2020); therefore, the five sampling sites were considered as replicates.

Despite the lack of seasonal effect in the baseline survey, in 2020 we repeated the sampling in winter (February–March) and summer (August–September) for comparative purposes. The 2020 summer samplings could not be conducted in June–July (as in 1995) because of the COVID-19 lockdown. Individual samples were placed into plastic bags, transported to the laboratory within the hour, and frozen at –20° C pending further processing. After thawing the samples, all decapods were sorted, identified with the highest taxonomic resolution possible using multiple sources (see Briones-Fourzán et al. 2020), quantified, and preserved in 70% ethanol.

Data analyses

The assemblage composition of decapods was analyzed using multivariate techniques with the *vegan* package (Oksanen et al. 2007) in R (R Core Team 2025). The abundance data were previously transformed to fourth-root to downplay the influence of highly abundant species, and the Bray-Curtis similarity measure was then calculated (Clarke 1993). Non-metric multidimensional scaling (nMDS) was used to visualize differences in the assemblage composition between Periods (2020, 1995), Seasons (winter, summer), and Times of day (day, night). The Bray-Curtis similarity matrix was used as an input to permutational multivariate analysis of variance (PERMANOVA) to test for the main effects of Period,

Season, and Time of day, two-way interactions between all pairs of these main effects, and the three-way interaction, all of which might be relevant to understanding the potential changes between the two periods (Potter et al. 2016). Permutational analyses of dispersion (PERMDISP) were also conducted using the same similarity matrix to determine whether there were differences in dispersion of samples between periods, seasons, and times of day (Warton et al. 2012). An analysis of similarity (ANOSIM) was then used to interpret the relative size of the overall Period, Season and Time of day effect on the decapod assemblage (Clarke and Warwick 2001). Finally, a similarity percentage analysis (SIMPER) (Clarke 1993) was conducted to identify those species responsible for the observed differences in assemblage composition. ANOSIM and SIMPER were done in the PRIMER v6 software (PRIMER-E Ltd) (Clarke and Gorley 2006).

For each sampling site, in addition to abundance (N), the following ecological indices were estimated by Period, Season, and Time of day: species richness (S , number of species), Shannon–Wiener’s diversity (H'), Pielou’s evenness (J'), and Simpson’s dominance ($1-\lambda$). S is the basis of biodiversity estimates, whereas the compound indices H' , J' and $1-\lambda$ provide a greater ability to discriminate samples because they combine richness and abundance of species (Magurran and Dornelas 2010; Morris et al. 2014). We used separate general linear models (GLM) to test the effects of Period, Season, Time of day, and their interactions on each ecological index. However, as noted by Bauer (1985a), differences in the abundance of individual species must also be considered, as data from all species together could be dominated by the few most numerous. Therefore, the data for eight of the most abundant species in both 2020 and 1995 were transformed using $\log(n+1)$ and subjected to separate GLM analyses. In all cases, Levene’s and Shapiro–Wilk’s tests were used to test for assumptions of homoscedasticity and normality. Significant GLM results were followed by a Tukey HSD test to identify the differences.

Species turnover, i.e. the change in species composition between 1995 and 2020, was quantified using Simpson’s dissimilarity, β_{Sim} (Simpson 1943), which is based on presence/absence and is independent of species richness (Koleff et al. 2003; Bianchi et al. 2014). This measure has been formulated as:

$$\beta_{\text{Sim}} = \frac{\min(b, c)}{\min(b, c) + a}$$

where b and c are the number of species unique to one or the other period and a is the number of species occurring in both periods. β_{Sim} can range from 0 (complete overlap, i.e. the same species composition) to 1 (no overlap, completely different species composition).

To appreciate the change of each species between periods, we used the index of relative dominance (I_{Rd}) developed by Parravicini et al. (2008), as formulated by Bianchi et al. (2014):

$$I_{\text{Rd}} = \frac{(N_{2020} - N_{1995})}{(N_{2020} + N_{1995})}$$

where N_{2020} is the raw abundance of a given species in 2020 and N_{1995} the raw abundance of the same species in 1995. I_{Rd} ranges from -1 (species found in 1995 only, i.e., ‘losses’) to $+1$ (species found in 2020 only, i.e., ‘gains’). Following Bianchi et al. (2014), arbitrary

limits were fixed for the values of I_{Rd} to determine those species that increased their abundance between periods ('winners': $I_{Rd} > 0.30$), those that reduced their abundance ('losers': $I_{Rd} < -0.30$); and those that underwent little or no change ('No change': $0.29 \leq I_{Rd} \leq -0.29$).

Results

Decapod epifauna in 2020

In 2020, we collected 138,045 decapod individuals in all, belonging to 108 taxa (mostly species). Of these individuals, 110,044 were identified to species level, 27,999 to genus level, and 2 to family level. Most of the individuals identified only to genus level were small carideans of the genus *Thor* ($n=7,680$ individuals) and hermit crabs of the genus *Pagurus* ($n=19,131$), which were damaged, missing appendages, or small juveniles. Therefore, for analytical purposes, these taxa were considered as *Thor* spp. and *Pagurus* spp. In addition to the latter, 7,719 individuals of *Pagurus* were identified to several species, mostly *P. brevidactylus* (Table 1).

Of the 108 taxa obtained in 2020, 21 were represented by only one individual and 6 by two individuals. Brachyura (43 species), Caridea (35 species), and Anomura (16 species) were the more diverse infraorders, followed by the superfamily Peneaoidea (suborder Dendrobranchiata, 11 species), and the infraorders Achelata (2 species) and Gebiidea (1 species). The most abundant infraorders were Anomura (61.6% of all individuals) and Caridea (35.1%), followed at a distance by Brachyura (2.3%), Penaeoidea (0.96%) and Achelata (0.001%), with Gebiidea represented by only one individual. The more diverse families were Palaemonidae (12 species), Diogenidae and Epialtidae (11 each), Mithracidae (10), Alpheidae (9), and Hippolythidae (8 species).

Comparisons between 2020 and 1995

Decapod species and abundance

The 138,045 decapods collected in 2020 represent a 2.5-fold increase over the total collected in 1995 ($n=53,211$) (Table 1). In both periods, most individuals belonged to relatively few taxa irrespective of season and time of day, as shown by the k-dominance curves, although these curves had a higher starting point along the y-axis in 2020, particularly the night samples (Fig. 2). Overall, in 2020, seven taxa (in decreasing order of abundance), *Clibanarius tricolor*, *Pagurus* spp., *Latreutes fucorum*, *Palaemonella americana*, *Thor* spp., *Pagurus brevidactylus*, and *Hippolyte zostericola*, accounted for nearly 90% of all individuals. For comparisons with these data, species of *Pagurus* (except *P. brevidactylus*) and *Thor* identified in 1995 were grouped as *Pagurus* spp. and *Thor* spp., respectively. Thus, in 1995, 90% of all individuals belonged to nine taxa (in decreasing order of abundance), *L. fucorum*, *P. americana*, *Thor* spp., *Pagurus* spp., *P. brevidactylus*, *C. tricolor*, *Alpheus normanni*, *Tozeuma carolinense*, and *Processa bermudensis* (Briones-Fourzán et al. 2020) (photos in Online Resource 2).

In general, the most abundant taxa reported in 1995 also remained among the most abundant in 2020, but the ranking order differed between periods (Table 1). In 1995, 64.4% of

Table 1 Total taxa and number of individuals of seagrass-associated decapods registered in the Puerto Morelos reef lagoon, Mexico

Family	Species/Lower taxon	Total 2020	Total 1995	I_{Rd}	Category
Diogenidae	<i>Clibanarius tricolor</i>	57,669	4641	0.85	Winner
Paguridae	<i>Pagurus</i> spp.	19,131	5803	0.53	Winner
Hippolythidae	<i>Latreutes fucorum</i>	17,801	11,005	0.24	No change
Palaemonidae	<i>Palaemonella americana</i>	9582	7634	0.11	No change
Thoridae	<i>Thor</i> spp.	7680	8445	-0.05	No change
Paguridae	<i>Pagurus brevidactylus</i>	7515	5193	0.18	No change
Hippolythidae	<i>Hippolyte zostericola</i>	3949	30	0.98	Winner
Hippolythidae	<i>Tozeuma carolinense</i>	2439	1528	0.23	No change
Processidae	<i>Processa bermudensis</i>	1624	1155	0.17	No change
Mithracidae	<i>Pitho lherminieri</i>	1077	25	0.95	Winner
Hippolythidae	<i>Hippolyte pleuracantha</i>	983	0	1.00	Gain
Alpheidae	<i>Alpheus normanni</i>	815	2037	-0.43	Loser
Processidae	<i>Processa fimbriata</i>	769	612	0.11	No change
Palaemonidae	<i>Leander tenuicornis</i>	549	657	-0.09	No change
Alpheidae	<i>Alpheus</i> sp.	507	8	0.97	Winner
Sicyonidae	<i>Sicyonia parri</i>	477	443	0.04	No change
Processidae	<i>Processa</i> sp.	467	0	1.00	Gain
Mithracidae	<i>Mithraculus coryphe</i>	450	100	0.64	Winner
Mithracidae	<i>Mithraculus sculptus</i>	447	178	0.43	Winner
Hippolythidae	<i>Trachycaris rugosa</i>	391	335	0.08	No change
Penaeidae	<i>Metapenaeopsis goodei</i>	368	293	0.11	No change
Hippolythidae	<i>Hippolyte obliquimanus</i>	326	6	0.96	Winner
Sicyonidae	<i>Sicyonia laevigata</i>	301	484	-0.23	No change
Mithracidae	<i>Mithraculus forceps</i>	260	180	0.18	No change
Paguridae	<i>Pagurus marshi</i>	220	0	1.00	Gain
Diogenidae	<i>Calcinus tibicen</i>	169	68	0.43	Winner
Processidae	<i>Nikoides schmitti</i>	160	138	0.07	No change
Epialtidae	<i>Chorinus heros</i>	159	98	0.24	No change
Palaemonidae	<i>Phycomenes siankaanensis</i>	140	0	1.00	Gain
Panopeidae	<i>Panopeus</i> sp.	122	3	0.95	Winner
Diogenidae	<i>Clibanarius antillensis</i>	119	30	0.60	Winner
Mithracidae	<i>Mithrax pleuracanthus</i>	110	57	0.32	Winner
Portunidae	<i>Achelous tumidulus</i>	101	230	-0.39	Loser
Panopeidae	<i>Panopeus occidentalis</i>	98	110	-0.06	No change
Paguridae	<i>Pagurus provenzano</i>	94	0	1.00	Gain
Penaeidae	<i>Metapenaeopsis smithi</i>	79	0	1.00	Gain
Diogenidae	<i>Paguristes puncticeps</i>	75	27	0.47	winner
Mithracidae	<i>Omalacantha bicornuta</i>	64	51	0.11	No change
Palaemonidae	<i>Gnathophyllum americanum</i>	63	24	0.45	winner
Portunidae	<i>Achelous ordwayi</i>	46	90	-0.32	Loser
Hippolythidae	<i>Latreutes parvulus</i>	37	97	-0.45	Loser
Pseudorhombilidae	<i>Scopoli</i> <i>nuttingi</i>	35	21	0.25	No change
Palinuridae	<i>Panulirus argus</i>	34	76	-0.38	Loser
Penaeidae	<i>Metapenaeopsis</i> sp.	32	0	1.00	Gain
Diogenidae	<i>Paguristes hernancortez</i>	32	0	1.00	Gain
Penaeidae	<i>Penaeus</i> sp.	32	0	1.00	Gain
Mithracidae	<i>Omalacantha antillensis</i>	30	0	1.00	Gain

Table 1 (continued)

Family	Species/Lower taxon	Total 2020	Total 1995	I_{Rd}	Category
Palaemonidae	<i>Urocaris longicaudata</i>	29	0	1.00	Gain
Inachidae	<i>Podochela macrodera</i>	28	52	-0.30	Loser
Palaemonidae	<i>Periclimenes yucatanicus</i>	26	0	1.00	Gain
Dromiidae	<i>Moreiradromia antillensis</i>	24	10	0.41	Winner
Epialtidae	<i>Macrocoeloma subparellellum</i>	23	18	0.12	No change
Alpheidae	<i>Alpheus armillatus</i>	23	3	0.77	Winner
Xanthidae	<i>Cataleptodius floridanus</i>	22	0	1.00	Gain
Palaemonidae	<i>Periclimenes</i> sp.	20	0	1.00	Gain
Epialtidae	<i>Macrocoeloma diplacanthum</i>	19	24	-0.12	No change
Calappidae	<i>Calappa gallus</i>	14	0	1.00	Gain
Epialtidae	<i>Macrocoeloma trispinosum</i>	13	1	0.86	Winner
Sicyonidae	<i>Sicyonia wheeleri</i>	13	0	1.00	Gain
Penaeidae	<i>Metapenaeopsis gerardoi</i>	13	0	1.00	Gain
Diogenidae	<i>Dardanus venosus</i>	10	53	-0.68	Loser
Calappidae	<i>Cryptosoma bairdii</i>	10	0	1.00	Gain
Diogenidae	<i>Paguristes anomalus</i>	9	0	1.00	Gain
Cryptochiridae	<i>Troglocarcinus corallicola</i>	9	0	1.00	Gain
Palaemonidae	<i>Periclimenes rathbunae</i>	9	0	1.00	Gain
Sicyonidae	<i>Sicyonia</i> sp.	8	0	1.00	Gain
Hippolythidae	<i>Hippolyte nicholsoni</i>	7	0	1.00	Gain
Inachoididae	<i>Stenorhynchus seticornis</i>	6	1	0.71	Winner
Diogenidae	Diogenidae 1	5	4	0.11	No change
Paguridae	<i>Iridopagurus reticulatus</i>	5	0	1.00	Gain
Epialtidae	<i>Macrocoeloma nodipes</i>	5	0	1.00	Gain
Majidae	<i>Thersandrus compressus</i>	4	11	-0.47	Loser
Epialtidae	<i>Macrocoeloma laevigatum</i>	4	4	0.00	Loss
Alpheidae	<i>Alpheus heterochaelis</i>	4	0	1.00	Gain
Pinnotheridae	<i>Glassella floridana</i>	4	0	1.00	Gain
Leucosiidae	<i>Acanthilia intermedia</i>	3	0	1.00	Gain
Pilumnidae	<i>Pilumnus caribaeus</i>	3	0	1.00	Gain
Palaemonidae	<i>Periclimenes patae</i>	3	0	1.00	Gain
Leucosiidae	<i>Uhlias limbatus</i>	3	0	1.00	Gain
Mithracidae	<i>Mithrax hispidus</i>	3	0	1.00	Gain
Diogenidae	<i>Petrochirus diogenes</i>	3	0	1.00	Gain
Mithracidae	<i>Teleophrys ruber</i>	2	9	-0.64	Loser
Xanthidae	<i>Paractaea nodosa</i>	2	0	1.00	Gain
Epialtidae	<i>Macrocoeloma</i> sp.	2	0	1.00	Gain
Penaeidae	<i>Penaeus aztecus</i>	2	0	1.00	Gain
Penaeidae	<i>Penaeus duorarum</i>	2	0	1.00	Gain
Lysmatidae	<i>Lysmata ankeri</i>	2	0	1.00	Gain
Palaemonidae	<i>Ancylomenes pedersoni</i>	1	535	-1.00	Loss
Diogenidae	<i>Paguristes tortugae</i>	1	142	-0.99	Loser
Alpheidae	<i>Alpheus armatus</i>	1	7	-0.75	Loser
Diogenidae	<i>Paguristes</i> sp.	1	3	-0.50	Loser
Epialtidae	<i>Epialtus longirostris</i>	1	3	-0.50	Loser
Upogebiidae	<i>Upogebia affinis</i>	1	1	0.00	No change
Scyllaridae	<i>Scyllarus americanus</i>	1	0	1.00	Gain

Table 1 (continued)

Family	Species/Lower taxon	Total 2020	Total 1995	I _{Rd}	Category
Mithracidae	<i>Mithrax tortugae</i>	1	0	1.00	Gain
Epialtidae	<i>Epialtus bituberculatus</i>	1	0	1.00	Gain
Leucosiidae	<i>Speloeophorus elevatus</i>	1	0	1.00	Gain
Xanthidae	<i>Paractaea rufopunctata</i>	1	0	1.00	Gain
Alpheidae	<i>Alpheopsis</i> sp.	1	0	1.00	Gain
Alpheidae	<i>Alpheus malleator</i>	1	0	1.00	Gain
Portunidae	<i>Achelous ventralis</i>	1	0	1.00	Gain
Leucosiidae	<i>Ebalia stimpsonii</i>	1	0	1.00	Gain
Alpheidae	<i>Alpheus paracrinitus</i>	1	0	1.00	Gain
Epialtidae	<i>Epialtus dilatatus</i>	1	0	1.00	Gain
Palaemonidae	<i>Typton</i> sp.	1	0	1.00	Gain
Palaemonidae	<i>Periclimenaeus</i> sp.	1	0	1.00	Gain
Epialtidae	<i>Macrocoeloma concavum</i>	1	0	1.00	Gain
Alpheidae	<i>Alpheus floridanus</i>	1	0	1.00	Gain
Mithracidae	<i>Mithrax</i> sp.	0	143	-1.00	Loss
Mithracidae	<i>Pitho aculeata</i>	0	131	-1.00	Loss
Panopeidae	<i>Eurypanopeus dissimilis</i>	0	15	-1.00	Loss
Panopeidae	<i>Neopanope packardii</i>	0	14	-1.00	Loss
Calappidae	<i>Calappa sulcata</i>	0	10	-1.00	Loss
Alpheidae	<i>Alpheus peasei</i>	0	9	-1.00	Loss
	Xanthoid G	0	7	-1.00	Loss
Porcellanidae	<i>Petrolisthes galathinus</i>	0	7	-1.00	Loss
Calappidae	<i>Cyclozodion angustum</i>	0	7	-1.00	Loss
Palaemonidae	<i>Tuleariocaris neglecta</i>	0	5	-1.00	Loss
Panopeidae	<i>Panopeus herbsti</i>	0	5	-1.00	Loss
	Xanthoid H	0	4	-1.00	Loss
	Xanthoid B	0	4	-1.00	Loss
	Xanthoid L	0	3	-1.00	Loss
Sicyonidae	<i>Sicyonia brevirostris</i>	0	3	-1.00	Loss
Sicyonidae	<i>Sicyonia stimpsoni</i>	0	3	-1.00	Loss
Panopeidae	<i>Hexapanopeus angustifrons</i>	0	3	-1.00	Loss
Euryplacidae	<i>Euryplax nitida</i>	0	3	-1.00	Loss
	Xanthoid F	0	2	-1.00	Loss
	Xanthoid A	0	2	-1.00	Loss
	Xanthoid M	0	2	-1.00	Loss
	Xanthoid I	0	2	-1.00	Loss
Mithracidae	<i>Thoe puella</i>	0	2	-1.00	Loss
Hippolythidae	<i>Hippolyte</i> sp.	0	2	-1.00	Loss
	Anomuran A	0	2	-1.00	Loss
	Xanthoid E	0	1	-1.00	Loss
	Xanthoid D	0	1	-1.00	Loss
	Xanthoid C	0	1	-1.00	Loss
	Xanthoid N	0	1	-1.00	Loss
	Xanthoid K	0	1	-1.00	Loss
	Xanthoid J	0	1	-1.00	Loss
Alpheidae	<i>Synalpheus fritzmuelleri</i>	0	1	-1.00	Loss
Alpheidae	<i>Synalpheus</i> sp. 3	0	1	-1.00	Loss

Table 1 (continued)

Family	Species/Lower taxon	Total 2020	Total 1995	I_{Rd}	Category
Alpheidae	<i>Synalpheus</i> sp. 2	0	1	-1.00	Loss
Alpheidae	<i>Synalpheus</i> sp. 1	0	1	-1.00	Loss
Stenopodidae	<i>Stenopus hispidus</i>	0	1	-1.00	Loss
Leucosiidae	<i>Speloeophorus pontifer</i>	0	1	-1.00	Loss
Porcellanidae	Porcellanid A	0	1	-1.00	Loss
Pilumnidae	<i>Pilumnus</i> sp.	0	2	-1.00	Loss
Grapsidae	<i>Pachygrapsus gracilis</i>	0	1	-1.00	Loss
Palaemonidae	<i>Periclimenes iridescens</i>	0	1	-1.00	Loss
	Majoid A	0	1	-1.00	Loss
Lysmatidae	<i>Lysmata anchisteus</i>	0	1	-1.00	Loss
Pilumnidae	<i>Lobopilumnus agassizii</i>	0	1	-1.00	Loss
Hippolythidae	<i>Latreutes inermis</i>	0	1	-1.00	Loss
Disciadidae	<i>Discias atlanticus</i>	0	1	-1.00	Loss
	Caridean A	0	2	-1.00	Loss
Alpheidae	<i>Alpheopsis trispinosa</i>	0	1	-1.00	Loss
Alpheidae	<i>Alpheopsis trigona</i>	0	1	-1.00	Loss
Epidaltidae	<i>Acanthonyx petiverii</i>	0	1	-1.00	Loss
Alpheidae	<i>Alpheus websteri</i>	0	1	-1.00	Loss

Samplings in 2020 were conducted using the same methodology as in the baseline survey in 1995. Taxa are ordered from higher to lower total abundance in 2020. I_{Rd} is an index of relative dominance that assesses changes in abundance of each species between periods. I_{Rd} ranges from -1 (taxa found only in 1995, i.e. 'losses' in column 'Category') to +1 (taxa found only in 2020, i.e. 'gains'). Other categories are 'winners', taxa that increased their abundance between periods ($I_{Rd} > 0.30$), 'losers', taxa that reduced their abundance ($I_{Rd} < -0.30$), and 'no change', taxa that underwent little or no change ($0.29 \leq I_{Rd} \leq -0.29$)

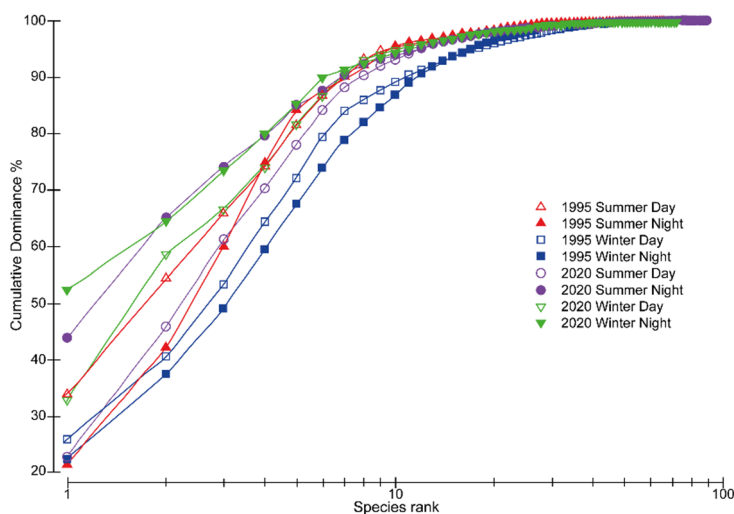


Fig. 2 k-Dominance curves for samples of seagrass-associated decapod crustaceans from the Puerto Morelos reef lagoon. Each curve is based on average abundances at five replicate sites. Samples from each site were obtained in 10 1-min trawls of an epibenthic sled in the day and 10 in the night, in the summer and winter of 1995 (baseline survey) and 2020

all collected decapods were Caridea and 30% were Anomura (Briones-Fourzán et al. 2020), whereas in 2020 the inverse occurred (Anomura: 61.6% of all individuals, Caridea: 35.1%). Most seagrass-associated decapods tend to be ‘benthic-dwellers’, i.e., stay on the bottom, remaining inactive (e.g. hermit crabs) or hiding in burrows under the sediment during the day (e.g. Alpheidae, Processidae). However, some are ‘canopy-dwellers’, i.e., cling to the seagrasses (e.g. *Latreutes fucorum*, *Hippolyte* spp., *Thor* spp.) or occur above the sediment surface as well as among seagrass leaves (e.g., *Palaemonella americana*, *Leander* spp.) (Bauer 1985a, b, 2023; Matheson et al. 1999). In 1995, ~50% of all individuals belonged to three taxa of canopy-dwelling caridean shrimps: *L. fucorum* (20.7%, $n=11,005$), *Thor* spp. (15.9%, $n=8,445$), and *P. americana* (previously *Cuapetes americanum*, 14.3%, $n=7,634$) (Briones-Fourzán et al. 2020) (Table 1). In 2020, by contrast, more than 50% of all individuals consisted of benthic-dwelling hermit crabs, *C. tricolor* (41.8%, $n=57,669$) and *Pagurus* spp. (13.9%, $n=19,131$), whereas *L. fucorum* and *P. americana* descended to third and fourth place, with 12.9% ($n=17,801$) and 6.9% ($n=9,582$) of all individuals, respectively (Table 1). Considering the absolute number of individuals, more *C. tricolor* were collected in 2020 than all decapod species combined in 1995, and even though *L. fucorum* and *P. americana* decreased in relative abundance, their absolute numbers were also higher in 2020 than in 1995.

Species turnover and changes in assemblage composition

Considering both periods, 159 decapod taxa were collected, of which 106 in 1995 and 108 in 2020. However, 52 occurred only in 1995 (‘losses’), 53 occurred only in 2020 (‘gains’), and 54 were common to both periods, yielding a turnover (β_{sim}) of 0.49. Most of the ‘losses’ involved relatively rare species or species identified only to genus or family level in 1995, with 1 to 15 individuals each, except for *Mithrax* sp. 1 and *Pitho aculeata*, which had 143 and 131 individuals, respectively. Similarly, many ‘gains’ in 2020 were species with less than 30 individuals, except for *Hippolyte pleuracantha*, *Processa* sp., and *Pagurus marshi*, which amounted to 983, 467, and 220 individuals, respectively (Table 1). Of the 53 species present in both periods, 18 were ‘winners’, with *C. tricolor*, *Pagurus* spp. and *H. zostericola* proportionally exhibiting the largest increases; 13 were ‘losers’, with *A. normanii*, *Achelous tumidulus* (identified as *Portunus* sp. in 1995) and *Paguristes tortugae* Schmitt, 1933 exhibiting the largest decreases, and 23 underwent little or ‘no change’ in abundance, most notably *L. fucorum*, *P. americana*, and *Thor* spp. (Table 1). Despite there being more taxa corresponding to ‘gains’ or ‘losses’ in 2020 (Fig. 3a), substantially more individuals belonged to taxa that were ‘winners’ and ‘no change’, than to ‘gains’, ‘losers’ or ‘losses’ (Fig. 3b).

Multivariate analyses were based on the abundance of each species by period, season and time of day (Online Resource 3). There were differences in the decapod assemblage composition between day and night samples, and between 1995 and 2020, but not between seasons (Fig. 4). The area of the convex hull of the 2020 samples in the nMDS plot was three times smaller (0.125 square units) than the convex hull of the 1995 samples (0.386) (Fig. 4).

Based on PERMANOVA, the decapod composition differed significantly with Period ($p<0.001$), Season ($p=0.026$) and Time of day ($p<0.001$), with a significant Period $\times$ Season interaction ($p=0.03$) (Table 2). Period and Time of day explained 21% and 18% of the variation, respectively, whereas Season explained only 3% (R^2 values in Table 2). However,

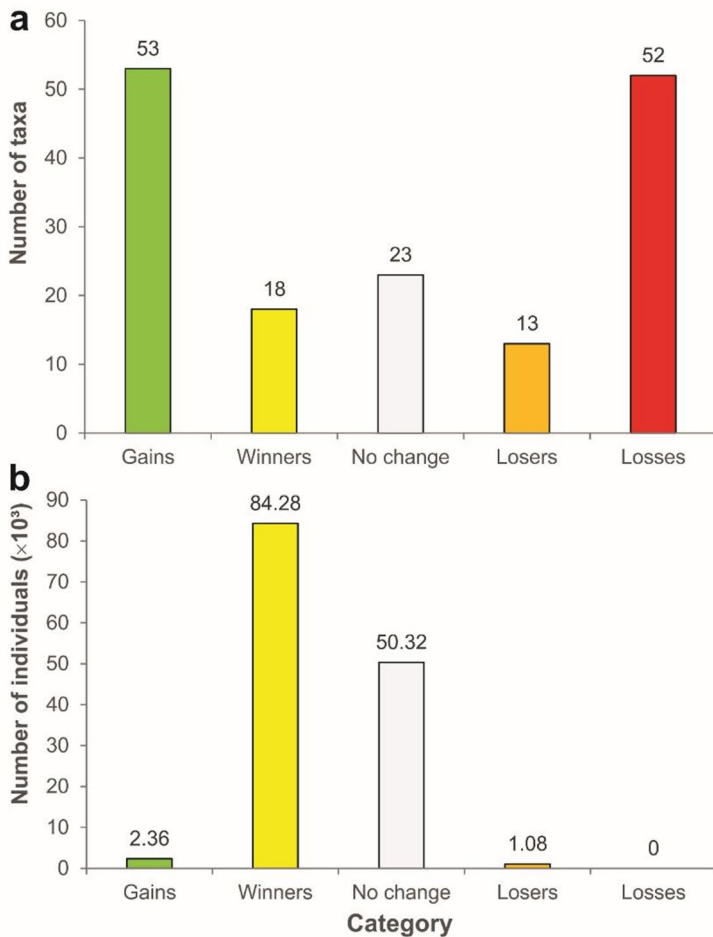


Fig. 3 (a) Number of taxa in each of five categories based on the value of the Index of Relative Dominance (I_{Rd}). ‘Gains’: taxa found only in 2020 ($I_{Rd} = 1$), ‘winners’: taxa that increased their abundance between 1995 and 2020 ($I_{Rd} > 0.30$), ‘No change’: taxa that underwent little or no change between 1995 and 2020 ($0.29 \leq I_{Rd} \leq -0.29$) ‘losers’: taxa that reduced their abundance between 1995 and 2020 ($I_{Rd} < -0.30$), and ‘losses’: taxa found only in 1995 ($I_{Rd} = -1$). (b) Number of individuals in each category

PERMDISP comparisons showed significant differences in dispersion between periods ($F_{1,38} = 6.54$, $p = 0.014$), with greater dispersion in 1995, and between times of day ($F_{1,38} = 10.944$, $p < 0.001$), with greater dispersion in day samples, but not between seasons ($F_{1,38} = 0.623$, $p = 0.4$). Also, there was a decrease in the distance of the centroids corresponding to the day samples between seasons in 2020 (0.242 Euclidean distances) relative to 1995 (0.401).

These results were confirmed by ANOSIM tests, which showed that the composition of the overall decapod assemblage varied significantly with period ($R = 0.530$, $p = 0.001$) and time of day ($R = 0.391$, $p = 0.001$), but not with season ($R = 0.036$, $p = 0.1$). When the combined effect of period and season was subjected to ANOSIM, the paired comparisons winter

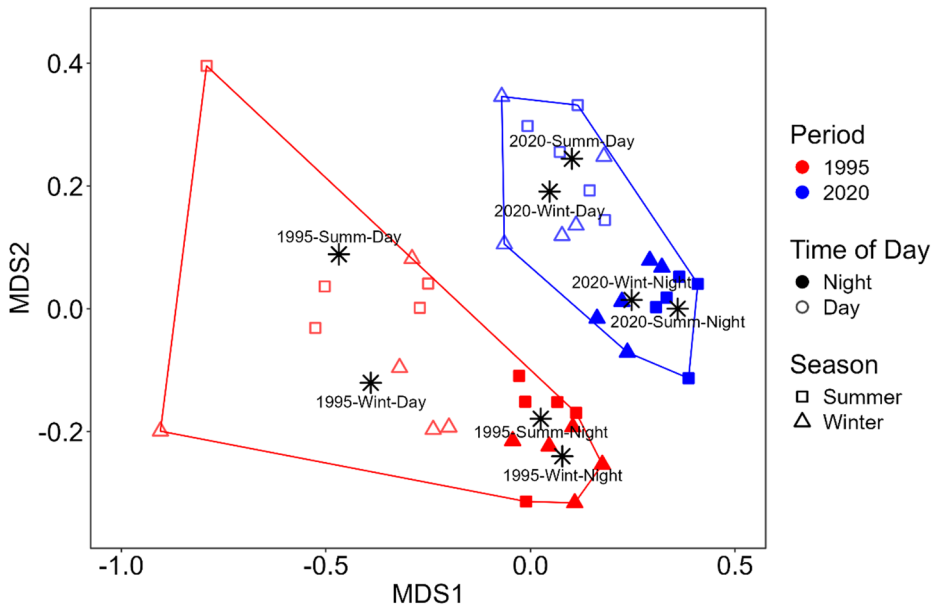


Fig. 4 Ordination analysis (non-metric multidimensional scaling, stress=0.16) of the community structure of seagrass-associated decapods in samples from Puerto Morelos reef lagoon, from day samples (open symbols) and night samples (filled symbols) taken in the summer (squares) and the winter (triangles) of 1995 (baseline survey, red) and 2020 (blue). Each symbol denotes a site. Convex hull areas are shown for all 1995 samples (red line) and all 2020 samples (blue line). Within each area (period), asterisks denote centroids for night and day samples taken in the summer and in the winter

Table 2 Results of the PERMANOVA test on the Bray-Curtis similarity matrix constructed for the abundance of seagrass-associated decapods in the Puerto Morelos reef lagoon in two periods (2020, 1995), two seasons (summer, winter) and two times of day (day, night) ($n=5$ sites per period, season, and time of day)

Effect	DF	MS	R^2	pF	p
Period	1	0.2131	0.213	13.757	<0.001
Season	1	0.0344	0.034	2.222	0.026
Time of Day (TD)	1	0.1784	0.178	11.514	<0.001
Period $\times$ Season	1	0.0309	0.031	1.998	0.030
Period $\times$ TD	1	0.0203	0.020	1.308	0.222
Season $\times$ TD	1	0.0154	0.015	0.996	0.415
Period $\times$ Season $\times$ TD	1	0.0116	0.012	0.750	0.660
Error	32	0.4958	0.496		
Total	39	3.7275	1.000		

Significant p -values are in bold

vs. summer 2020 ($R=0.11$, $p=0.1$) and winter vs. summer 1995 ($R=0.08$, $p=0.1$) were also not significant, but the comparisons by season between periods was significant (winter 2020 vs. 1995: $R=0.544$, $p<0.001$; summer 2020 vs. 1995: $R=0.571$, $p<0.001$). When the combined effect of period and time of day was analyzed, the day vs. night comparison was significant ($p<0.001$) in both 2020 ($R=0.528$) and 1995 ($R=0.565$), but the differences were even greater in day samples between 2020 and 1995 ($R=0.638$), and in night samples between 2020 and 1995 ($R=0.786$). Therefore, period and time of day had a greater effect than season on decapod assemblage composition, but the significant differences found

between 2020 and 1995 and between day and night samples were driven by both location and dispersion effects (Warton et al. 2012).

Given the small effect of season, we ran SIMPER analysis on the combined factors period and time of day (Online Resource 4). Similarity was high among day samples and among night samples within each period. Dissimilarity between day and night samples was lower in 2020 (37.4%) than in 1995 (46.4%), while dissimilarity between periods was greater for day (48.4%) than for night samples (38.9%). Within each period, dissimilarity between day and night samples reflected the higher abundance of most taxa at night, but in 1995 the three species most contributing to this dissimilarity were *P. bermudensis*, *A. normanni*, and *C. tricolor*, whereas in 2020 they were *C. tricolor*, *P. bermudensis*, and *Pagurus* spp. Between periods, three species, *C. tricolor*, *H. zostericola*, and *Pitho lherminieri*, contributed most to dissimilarity within both day and night samples. All three species were much more abundant during the day and night in 2020 than in 1995 (Online Resource 4).

Ecological indices

Levene's and Shapiro-Wilk's tests revealed that data from four of the five estimated ecological indices met the GLM assumptions. The exception was the raw abundance data (N), but the fourth-root transformation did. Species richness (S) and abundance ($\sqrt[4]{N}$) varied significantly with Period (higher values in 2020) and Time of day (higher values at night), but not with season, although the only significant interaction was Period $\times$ Season (Table 3), with higher values of both S and $\sqrt[4]{N}$ in summer 2020 than in winter 2020, and summer and winter of the baseline survey (Fig. 5). Evenness (J'), diversity (H'), and dominance ($1-\lambda$) were not affected by Period, Season, Time of day, or any interaction (Table 3; Fig. 5).

Changes in abundance of eight of the most abundant species in 2020 and 1995

Period, Season and Time of day had no effect on the abundance of *P. americana*, *T. carolinense*, and *L. fucorum*, although the Period $\times$ Season interaction was significant for the latter (Table 4), with lower values in the summer of 2020 than in the other three seasons; however, intra-group comparison revealed no significant difference. Period had a significant effect only on the abundance of *C. tricolor*, *Pagurus* spp. and *P. brevidactylus*, with much higher values of all three taxa in 2020 than in 1995. The abundance of the latter two taxa was also significantly greater at night, and in the case of *Pagurus* spp., also in the summer. This was also the case for *P. bermudensis*, whose abundance was higher in the summer and at night. The abundance of *Thor* spp. only varied significantly with time of day, with higher values at night.

Discussion

Our results revealed that the abundance of seagrass-associated decapods in the Puerto Morelos reef lagoon increased 2.5 times in 2020 compared to 1995, i.e., after 25 years of gradually changing conditions, from oligotrophic to mesotrophic-eutrophic. Although the total number of taxa was similar in both periods, there was a nearly 50% species turnover, with most of the 'losses' and 'gains' consisting of taxa with relatively low abundances. Most taxa

Table 3 Results of general linear models ($\alpha=0.05$) on data of five ecological indices of decapod diversity (S : taxa richness; $\sqrt[4]{N}$: fourth-root abundance; H' : Shannon-Wiener's diversity; J' : pielou's evenness; $1-\lambda$: simpson's dominance) in two periods (2020, 1995), two seasons (summer, winter) and two times of day (day, night) in seagrass meadows of the Puerto Morelos reef lagoon ($n=5$ sites)

Ecological index	Effect	DF	MS	F	<i>p</i>
S	Intercept	1	66912.4	1423.29	<0.001
	Period	1	756.9	16.1	<0.001
	Season	1	16.9	0.359	0.553
	Time of Day (TD)	1	940.9	20.014	<0.001
	Period $\times$ Season	1	336.4	7.156	0.012
	Period $\times$ TD	1	25.6	0.545	0.466
	Season $\times$ TD	1	1.6	0.034	0.855
	Period $\times$ Season $\times$ TD	1	72.9	1.551	0.222
	Error	32	47.01		
$\sqrt[4]{N}$	Intercept	1	263737.6	1464.192	<0.001
	Period	1	6051.6	33.597	<0.001
	Season	1	547.6	3.040	0.091
	Time of Day (TD)	1	10758.4	59.727	<0.001
	Period $\times$ Season	1	1440.0	7.994	0.008
	Period $\times$ TD	1	129.6	0.720	0.403
	Season $\times$ TD	1	57.6	0.320	0.576
	Period $\times$ Season $\times$ TD	1	313.6	1.741	0.196
	Error	32	180.1		
H'	Intercept	1	168.81	945.313	<0.001
	Period	1	0.1292	0.723	0.401
	Season	1	0.2216	1.241	0.274
	Time of Day (TD)	1	0.0125	0.070	0.793
	Period $\times$ Season	1	0.4531	2.537	0.121
	Period $\times$ TD	1	0.0873	0.489	0.489
	Season $\times$ TD	1	0.0446	0.250	0.621
	Period $\times$ Season $\times$ TD	1	0.0004	0.002	0.964
	Error	32	0.1786		
J'	Intercept	1	12.4960	1181.447	<0.001
	Period	1	0.0436	4.125	0.051
	Season	1	0.0186	1.758	0.194
	Time of Day (TD)	1	0.0225	2.130	0.154
	Period $\times$ Season	1	0.0140	1.325	0.258
	Period $\times$ TD	1	0.0009	0.086	0.771
	Season $\times$ TD	1	0.0044	0.412	0.525
	Period $\times$ Season $\times$ TD	1	0.0013	0.123	0.729
	Error	32	0.0106		
$1-\lambda$	Intercept	1	24.1262	1758.625	<0.001
	Period	1	0.0168	1.224	0.277
	Season	1	0.0021	0.155	0.696
	Time of Day (TD)	1	0.0077	0.563	0.459
	Period $\times$ Season	1	0.0107	0.777	0.387
	Period $\times$ TD	1	0.0082	0.594	0.447
	Season $\times$ TD	1	0.0180	1.311	0.261
	Period $\times$ Season $\times$ TD	1	0.0014	0.105	0.748
	Error	32	0.0137		

Significant *p*-values are in bold

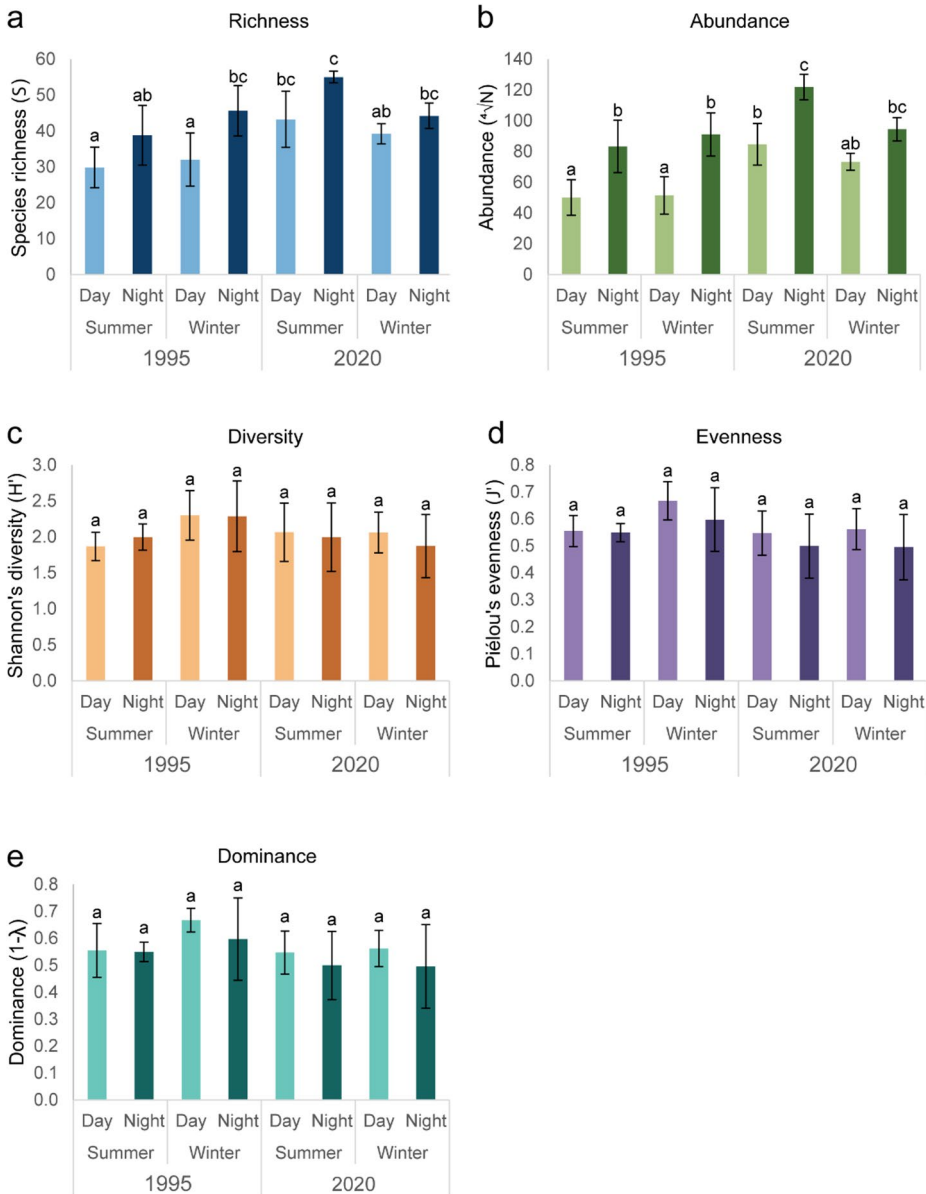


Fig. 5 Mean ecological indices of decapod diversity. **(a)** S : taxa richness, **(b)** $\sqrt[4]{N}$: fourth-root abundance, **(c)** H' : Shannon–Wiener’s diversity, **(d)** J' : Pielou’s evenness, **(e)** $1-\lambda$: Simpson’s dominance, estimated in seagrass meadows of the Puerto Morelos reef lagoon ($n=5$ sites) during the day (lighter columns) and night (darker columns) in two seasons (summer, winter) of 1995 (baseline survey) and 2020. In each panel, the same letter indicates similar means ($\alpha=0.05$). Error bars are 95% confidence intervals

Table 4 *p*-values from individual general linear models ($\alpha=0.05$) to test for the effects of period (2020, 1995), season (winter, summer), time of day (day, night), and all their interactions on the abundance ($\log(n+1)$) of each of eight of the most abundant taxa in both 2020 and 1995

Taxon	Period (P)	Season (S)	Time of day (TD)	$P \times S$	$P \times TD$	$S \times TD$	$P \times S \times TD$
<i>Clibanarius tricolor</i>	<0.001	0.992	0.068	0.928	0.946	0.956	0.788
<i>Pagurus</i> spp.	0.002	0.003	<0.001	0.401	0.651	0.178	0.480
<i>Latreutes fucorum</i>	0.213	0.215	0.968	0.030	0.062	0.852	0.280
<i>Palaemonella americana</i>	0.208	0.093	0.054	0.594	0.190	0.810	0.974
<i>Thor</i> spp.	0.195	0.383	0.022	0.728	0.596	0.734	0.689
<i>Pagurus brevidactylus</i>	0.009	0.109	<0.001	0.125	0.471	0.538	0.444
<i>Tozeuma carolinense</i>	0.195	0.948	0.188	0.206	0.383	0.354	0.914
<i>Processa bermudensis</i>	0.055	0.014	<0.001	0.229	0.364	0.137	0.773

Significant *p*-values are in bold

that were abundant in 1995 persisted in 2020, although their rankings changed. Notably, the three most abundant taxa in 1995 were canopy-dwelling caridean shrimps, whereas in 2020 the most abundant taxon was the diogenid hermit crab *C. tricolor*, a ‘winner’ species with more individuals than the number of all decapods collected in 1995, followed by pagurid hermit crabs (*Pagurus* spp.). Other diogenids, such as *Calcinus tibicen*, *Clibanarius antillensis*, and *Paguristes puncticeps*, were also ‘winners’, although their numbers were much lower than those of *C. tricolor*.

Benthic communities may respond differently to eutrophication depending on the type of environment or the amount of organic matter reaching the sediments (Heip 1995; Jessen et al. 2015). For example, in an estuary in Portugal, eutrophication caused marked changes in macroinvertebrate assemblages, including declines in species diversity, increases in detritivores, and declines in herbivores (Cardoso et al. 2004). In the shallow Gulf of Batabanó, Cuba, benthic invertebrates underwent a decline in density and biomass with increasing habitat degradation due partly to eutrophication (Lopeztegui-Castillo and Martínez-Coello 2020). In contrast, in a microtidal coastal system in Patagonia, Argentina, eutrophication increased food availability for consumers, enhancing the abundance of benthic invertebrates (Martinetto et al. 2010). In coastal areas of Brazil subjected to eutrophication, an increase in the cover of brown and green macroalgae resulted in an increase in the abundance of herbivore mithracid crabs (Giraldes et al. 2021), whereas in a coral reef in Barbados, crustacean fauna responded mainly to eutrophication by a reduction in the density of individuals, but with no changes in species richness or number of rare species (Snelgrove and Lewis 1989). In Florida Bay, following extensive seagrass die-offs, declines occurred among seagrass canopy-dwelling species and increases among benthic seagrass-associated crustaceans, but species richness increased at most sites (Matheson et al. 1999). In the Puerto Morelos reef lagoon, differences in the decapod community composition related to eutrophication between 1995 and 2020 included a change in numerical dominance from caridean shrimps to small hermit crabs, particularly *C. tricolor*.

Although *C. tricolor* is common in reef systems throughout the wider Caribbean, it shows wide fluctuations in its abundance. For example, Hazlett (1966) noted that it was the most abundant hermit crab in his samples from Curaçao, even though it had not been previously reported on the island. Baeza and Behringer (2017) reported substantial small-scale variation in the abundance of *C. tricolor* over a 100 km stretch of the Florida Keys, USA.

On a coral reef in Puerto Morelos, where *C. tricolor* was not recorded in 2015 (González-Gómez et al. 2018), aggregations exceeding 5000 individuals were observed in 2022 (Dubé et al. 2023). *C. tricolor* was not found among the decapods associated with sea anemones in the Puerto Morelos reef lagoon in 2007–2008 (Briones-Fourzán et al. 2012), but by 2016, it was observed forming clusters of dozens along the columns of sun anemones (*Stichodactyla helianthus*) in Akumal, a highly degraded bay ~70 km south of Puerto Morelos (Colombara et al. 2017). These findings suggest that the abundance of *C. tricolor* has been increasing for several years. Similarly, at Carrie Bow Cay, Belize, aggregations of up to 400 hermit crabs, including *C. tricolor* and pagurids, were recorded in association with seagrass anemones, mainly *S. helianthus*, in 2018 (Brooker et al. 2019).

Fluctuations in the abundance of small hermit crabs may reflect changes in the resources they use, because these crabs are highly site-attached and their movement ranges are very limited (Hazlett 1983; Bauer 1985b). Hermit crabs are opportunistic omnivores, but mostly mesograzers (*Pagurus*: Ruesink 2000; *Paguristes*: Petersen et al. 2005; Baggett et al. (Baggett, et al., 2013)) or deposit feeders (i.e. detritivores) (Hazlett 1981; Schembri 1982; Volvenko 1994), in particular *Clibanarius* species (Kunze and Anderson 1979; Hazlett 1983). Deposit feeders consume organic detritus derived from sediment particles of any size or from irregularities in the surface of larger objects (Schembri 1982). In 1999, when the Puerto Morelos reef lagoon was still an oligotrophic system, the organic matter content in the sediments was low (1.5–4.6%, 2% on average, Enríquez et al. 2001). In 2015, when extensive *Sargassum* brown tides occurred for the first time, there was a 15 to 35-fold increase in organic matter content up to 200 m from the coastline into the reef lagoon (van Tussenbroek et al. 2017). Over time, however, the impacts of the *Sargassum* brown tides propagate across the seagrass landscape to the coral reefs (Cabanillas-Terán et al. 2019; Velázquez-Ochoa and Enríquez 2023; Ávila-Mosqueda et al. 2025). The reef lagoon is generally well flushed, with a water turnover time of 2.5 h on average, but the water residence time can increase markedly, up to 2 weeks, during extended periods of low-energy waves (Coronado et al. 2007). Under these conditions, the decomposing *Sargassum* can remain in the system for long periods (Rutten et al. 2021), and this allochthonous organic matter becomes trapped because seagrasses further attenuate flow velocity (Agawin and Duarte 2002; Peterson et al. 2004). By 2018, the organic matter content in sediments throughout the reef lagoon had increased substantially (up to $8.01 \pm 0.29\%$, mean $\pm$ SE, in some areas), mainly due to anthropogenic sources but exacerbated by *Sargassum* brown tides (Velázquez-Ochoa and Enríquez 2023). The large and significant increase in abundance of *C. tricolor* and pagurids in 2020 compared with 1995 appears consistent with the increase in organic matter content in the sediments of the shallow Puerto Morelos reef lagoon.

Despite the change in overall numerical dominance from caridean shrimps to hermit crabs, the former also continued to be abundant. For example, even though the relative abundance of the most abundant carideans in 1995 (*L. fucorum*, *P. americana*, *Thor* spp., and *T. carolinense*) decreased in 2020, their absolute numbers increased somewhat, but did not vary significantly between periods, resulting in ‘No change’ in their indices of relative importance. Also, the number one ‘gain’ in 2020 was *H. pleuracantha*, whereas the ‘winner’ with the highest I_{rd} (0.98) was *H. zostericola*, followed shortly by *H. obliquimanus* (0.96). Hippolytids and thorids, as well as seagrass-associated palaemonids like *Palaemonella*, feed on seagrass epiphytes and occasionally consume smaller animals such as amphipods, which also feed on epiphytes (Zupo and Nelson 1999; Barba et al. 2000; Quiñones-Rivera and

Fleeger 2005; Baggett et al. 2013; Bauer 2023). Increased nutrient loads enhance epiphyte growth (Morris et al. 2007; Kelaher et al. 2013) and mediate changes in the leaf morphology of *T. testudinum*, hence in the habitat value for seagrass-associated invertebrates (Barry et al. 2021). In Puerto Morelos, eutrophication has led to a shift toward relatively higher biomass invested in above-ground tissues by seagrasses (Rodríguez-Martínez et al. 2010; van Tussenbroek 2011) and denser seagrass canopies (Hedley et al. 2021; de Almeida et al. 2022), which may favor canopy-dwelling species. Nutrient increases have also enhanced fleshy algal growth (van Tussenbroek 2011; van Tussenbroek et al. 2014), potentially favoring herbivore mithracid crabs (Giraldes et al. 2021). Although mithracid crabs were not among the most abundant species in the Puerto Morelos reef lagoon, several were among the ‘winner’ species in 2020 (e.g. *Pitho lherminieri*, *Mithraculus coryphe*, *M. sculptus*, and *Mithrax pleuracanthus*).

The significant period $\times$ season interaction for both mean taxa richness (S) and abundance (N), which showed higher values in the summer of 2020 relative to 1995, could partially reflect that the 2020 summer sampling was conducted two months later than in 1995 due to the COVID-19 lockdown. The higher values of mean S and N in night vs. day samples, and the difference in day vs. night decapod assemblage composition, had been previously reported in this reef lagoon by Briones-Fourzán et al. (2020) and is consistent with the nocturnal habits of most seagrass-associated decapods (Bauer 1985a, b; Howard 1987; de Grave et al. 2006; García-Trasviña et al. 2023). However, the decapod assemblage became more similar between day and night samples in 2020 compared with 1995. This may reflect the differential effect of Time of day on the abundance of different species (Bauer 1985a). Indeed, although the abundance of pagurids and processids was significantly higher at night, the abundance of *C. tricolor*, *P. americana*, and the hippolytids *L. fucorum* and *T. carolinense* did not differ between day and night, whereas *H. pleuracantha* and *H. obliquimanus* were much more abundant in day vs. night samples in 2020. Hippolytids rest during the day on the leaves of seagrasses (Bauer 1985a) and thus may be vulnerable to the epibenthic sled irrespective of time of day.

Conclusions

In tropical seagrass ecosystems, increases in nutrient inputs first enhance the biomass of both seagrasses and epiphytes and increase the organic matter content in sediments, but over time, may limit light availability, at which point seagrasses decline and macroalgae increase (Burkholder et al. 2007). As noted before, how benthic faunal communities respond to eutrophication may depend on the type of environment or the amount of organic matter reaching the sediments (Jessen et al. 2015). The empirical model of Pearson and Rosenberg (1978) predicts three successive stages in benthic communities in response to increasing organic matter inputs to the sediments: (1) few or no changes in species composition and slight increases in biomass, (2) replacement of ‘normal’ species by opportunistic species and strong increases of abundance and biomass, and (3) disappearance of benthic animal species and sediments with no fauna (Heip 1995; Grall and Chauvaud, 2002). The changes in the seagrass-associated decapod assemblage in the Puerto Morelos reef lagoon between 1995, when the reef lagoon was oligotrophic, and 2020, when it had become mesotrophic-eutrophic, mainly consisted of a substantial increase in the number of individuals and a shift

in dominance from canopy-dwelling, omnivorous/herbivorous caridean shrimps to small bottom-dwelling, omnivorous/detritivorous hermit crabs, as well as a shift in assemblage composition. These changes would suggest that by 2020 the benthic communities in this shallow reef lagoon were reaching the second stage of Pearson and Rosenberg's (1978) model compared with the baseline survey of 1995 (Briones-Fourzán et al. 2020).

Our results provide insights into how ecological changes are affecting seagrass-associated invertebrates in the Caribbean region, where coastal eutrophication is on the rise due to human activities and continuous *Sargassum* inundations (van Tussenbroek et al. 2014; UNEP-CEP 2021). They also suggest that seagrass-associated decapod assemblages dominated by hermit crabs might be indicative of benthic eutrophication in other Caribbean seagrass ecosystems.

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Data availability Data is provided within the manuscript or supplementary information files.

Declarations

Competing interests The authors declare no competing interests.

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